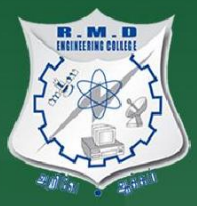
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R.M.K. ENGINEERING COLLEGE

(An Autonomous Institution)

DEPARTMENT OF INFORMATION TECHNOLOGY

22CS603 - PROFESSIONAL ETHICS

Department : IT
Batch/Year : 2022-2026 / VI
Created by : Dr. M. Sheerin Banu
Ms. S. Nandhini
Date : 02.07.2026

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2. COURSE OBJECTIVES

The Course will enable learners to:

- **Familiarize with Engineering Ethics and Human Values.**
- **Impart knowledge on codes of ethics, safety, responsibilities and rights of engineers.**
- **Give awareness on global issues related to environmental**

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3. PRE REQUISITES

- **Ethics in engineering practice is about professional responsibilities of engineers.**
- **Professional ethics have been recognized as an important foundation in the practice of engineering for several decades in many industrialized countries.**



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4. Syllabus

22CS603- PROFESSIONAL ETHICS

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3	0	0	3

UNIT I HUMAN VALUES 10

Morals, values and Ethics – Integrity – Work ethic – Service learning – Civic virtue – Respect for others – Living peacefully – Caring – Sharing – Honesty – Courage – Valuing time – Cooperation - Commitment – Empathy – Self confidence – Character – Spirituality – Introduction to Yoga and meditation for professional excellence and stress management.

UNIT II ENGINEERING ETHICS 9

Senses of ‗Engineering Ethics – Variety of moral issues – Types of inquiry – Moral dilemmas –Moral Autonomy – Kohlberg‘s theory – Gilligan‘s theory – Consensus and Controversy – Models of professional roles - Theories about right action – Self-interest – Customs and Religion – Uses of Ethical Theories.

UNIT III ENGINEERING AS SOCIAL EXPERIMENTATION 9

Engineering as Experimentation – Engineers as responsible Experimenters – Codes of Ethics – A Balanced Outlook on Law.

UNIT IV SAFETY, RESPONSIBILITIES AND RIGHTS 9

Safety and Risk – Assessment of Safety and Risk – Risk Benefit Analysis and Reducing Risk

- Respect for Authority – Collective Bargaining – Confidentiality – Conflicts of Interest – Occupational Crime – Professional Rights – Employee Rights – Intellectual Property Rights (IPR) – Discrimination.

UNIT V GLOBAL ISSUES 8

Multinational Corporations – Environmental Ethics – Computer Ethics – Weapons Development – Engineers as Managers – Consulting Engineers – Engineers as Expert Witnesses and Advisors – Moral Leadership –Code of Conduct – Corporate Social Responsibility.

5. COURSE OUTCOMES

CO	Course outcome(CO) – Statements
CO1	Summarize the importance of human values in work place.
CO2	Discuss the senses of engineering ethics, moral dilemmas, moral autonomy and uses of ethical Theories
CO3	Describe the role of engineers as responsible experimenters and necessity of codes of ethics in engineering.
CO4	Explain safety, risk, responsibilities and rights in the society.
CO5	Analyze the global issues related to environmental ethics, computer ethics, weapons development and the role of engineers as expert witnesses and advisors.
CO6	Apply ethics in society and discuss the ethical issues related to engineering.

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COURSE OUTCOMES WITH KNOWLEDGE LEVEL AND JUSTIFICATION

CO. NO	COURSE OUTCOMES	KNOWLEDGE LEVEL	CO JUSTIFICATION
CO1	Summarize the importance of human values in work place.	K1	Understand the personal concepts of responsibility, entitlement and respect. Describe how the individuals interact ethically with colleagues.
CO2	Discuss the senses of engineering ethics, moral dilemmas, moral autonomy and uses of ethical theories.	K3	Explain moral ideals, character, policies and relationships of people
CO3	Describe the role of engineers as responsible experimenters and necessity of codes of ethics in engineering.	K1	Analyze how learning from the past mistakes helps to overcome the problems
CO4	Explain safety, risk, responsibilities and rights in the society.	K1	Understand the need of awareness of its risk. Identify the rights of customers or clients.
CO5	Analyze the global issues related to environmental ethics, computer ethics, weapons development and the role of engineers as expert witnesses and advisors.	K1	Understand an MNCs Culture and its expansion in other countries.
CO6	Apply ethics in society and discuss the ethical issues related to engineering.	K1	Understand the need of engineering ethics.

6. CO- PO/PSO Mapping

Course Outcomes	Level of CO	Program Outcomes												Program Specific Outcomes		
		PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	K2	-	-	-	-	-	-	-	2	1	1	-	3	-	-	-
CO2	K2	-	-	-	-	-	2	3	2	1	1	-	3	-	-	-
CO3	K2	-	-	-	-	-	2	3	2	1	1	1	2	-	-	-
CO4	K2	-	-	-	-	-	2	2	2	1	1	-	3	-	-	-
CO5	K2	-	-	-	-	-	2	1	2	2	1	1	2	-	-	-
CO6	K2	-	-	-	-	-	2	2	2	-	1	-	3	-	-	-
CO		-	-	-	-	-	2	2	2	1	1	1	3	-	-	-

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

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UNIT V GLOBAL ISSUES

Multinational Corporations – Environmental Ethics – Computer Ethics – Weapons Development – Engineers as Managers – Consulting Engineers – Engineers as Expert Witnesses and Advisors – Moral Leadership –Code of Conduct – Corporate Social Responsibility

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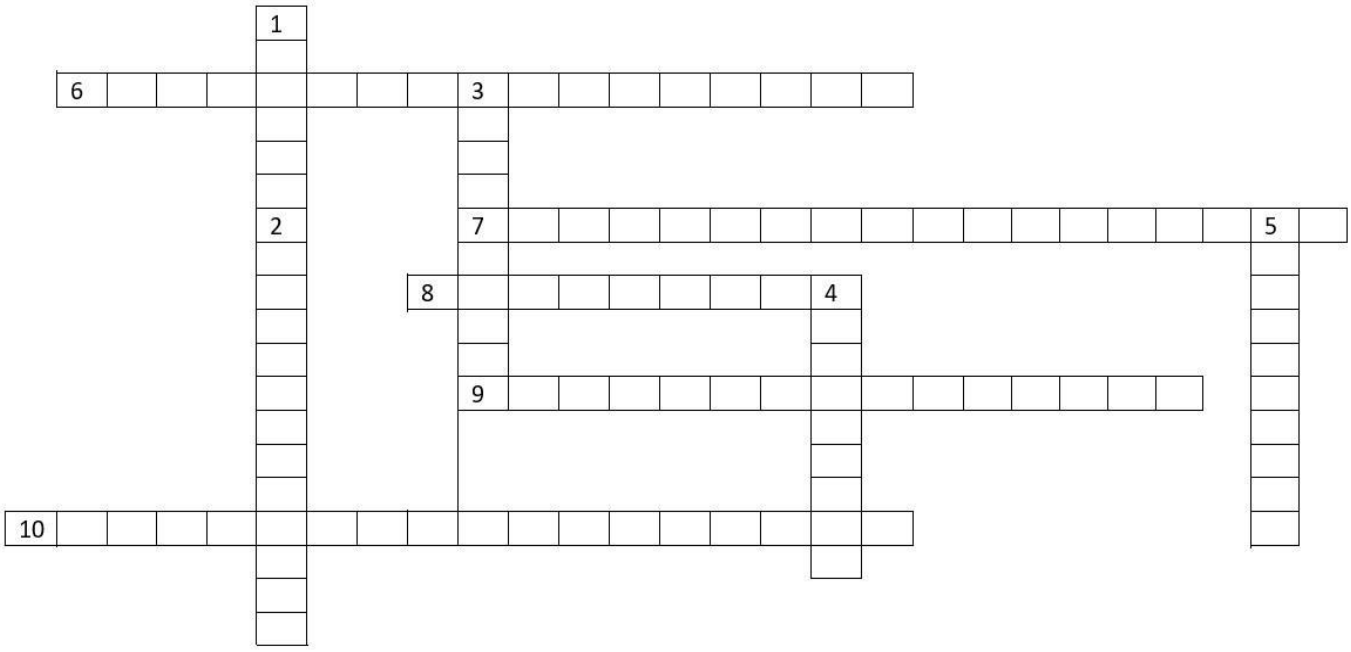
7. Lecture Plan

UNIT V GLOBAL ISSUES

S.No	Topic	No. of Period	Proposed Date	Actual Date	Pertaining CO	Taxonomy level	Mode of Delivery	Reason for Deviatio
1	Multinational Corporations	1			CO5	K2 Understand	PowerPoint through online	
2	Environmental Ethics	1			CO5	K2 Understand	PowerPoint through online	
3	Computer Ethics	1			CO5	K2 Understand	PowerPoint through online	
4	Weapons Development	1			CO5	K2 Understand	PowerPoint through online	
5	Engineers as Managers – Consulting Engineers	1			CO6	K2 Understand	PowerPoint through online	
6	Engineers as Expert Witnesses and Advisors	1			CO6	K2 Understand	PowerPoint through online	
7	Moral Leadership	1			CO6	K2 Understand	PowerPoint through online	
8	Code of Conduct	1			CO6	K1 Remembrance	PowerPoint through online	
9	Corporate Social Responsibility	1			CO6	K1 Remembrance	PowerPoint through online	

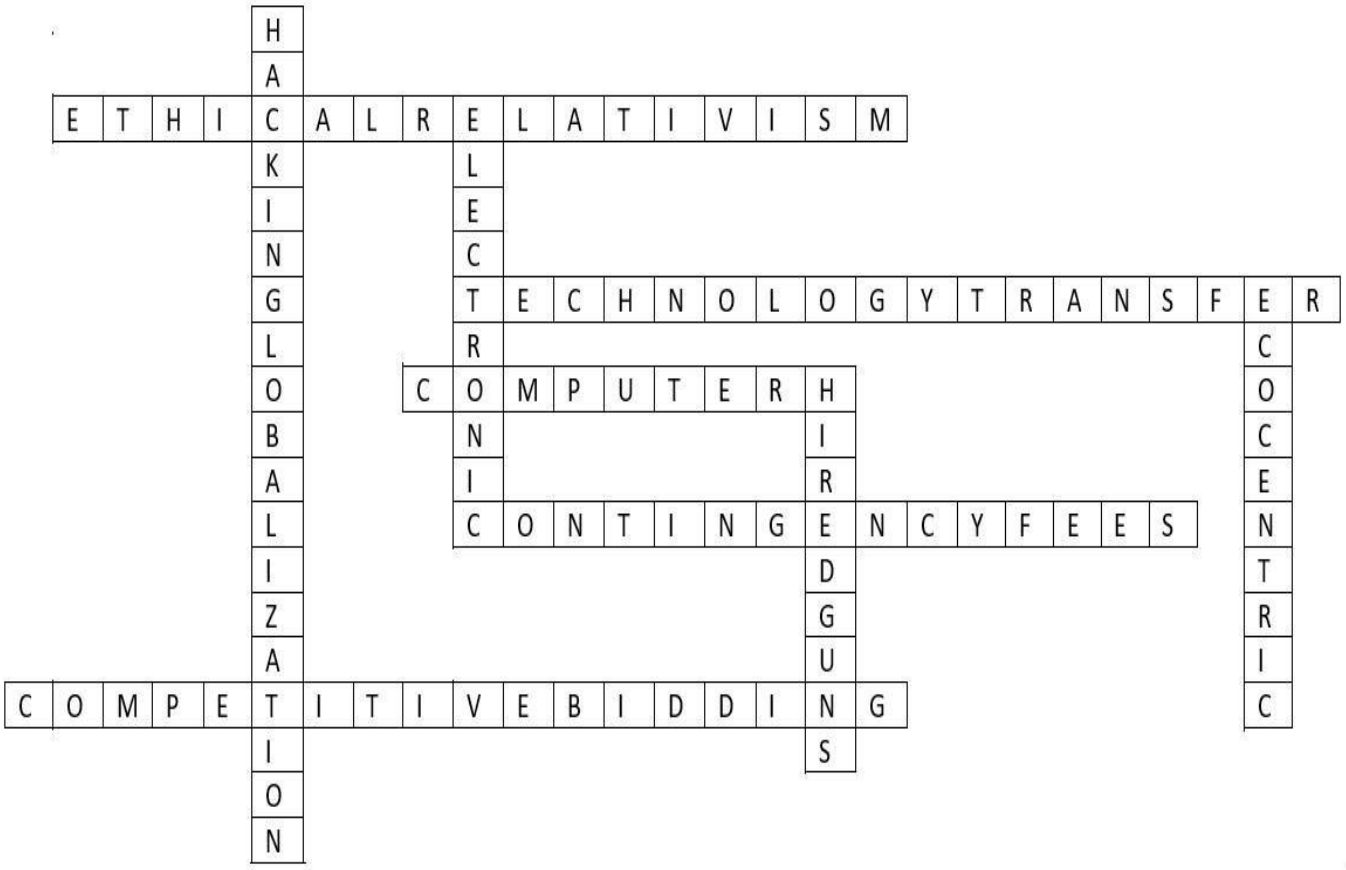
Total No. of Periods : 9

8. ACTIVITY BASED LEARNING
CROSSWORD PUZZLE



Down	Across
1._____is nothing but gaining unauthorized access to database.	6.Which says that actions are morally right when they are approved by law or custom they are wrong when they violate
2.The process of interaction and integration among people, companies and government worldwide is called ?	7._____is the process of moving technology to a quite new set of conditions and implementing it there.
3.What is e- waste ?	8.The study of ethical issues that are associated primarily with computing machines and computing profession is ?
4.When do engineers are hired by attorneys to help them to establish the facts in a way favorable to their clients, then they become?	9._____can be paid only for those consultant engineers who succeed in profiting or saving the client money.
5._____ethics emphasizes inherent values in ecological systems.	10.The process of offering of price at an auction or in business to achieve something is ?

Answer



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CASE STUDY

ENVIRONMENT CONCERN

You are an environmental engineer for one of the many local plants. That plant discharges effluents into a lake in a flourishing tourist area. Although all the plants are marginally profitable, they compete for the same customers. Your responsibilities are to monitor the water and air discharges at your plant and the periodic reporting to Dept. of Anti-pollution. You have just prepared a report that indicates that the level of pollution in the Plant's water discharges slightly exceed the legal limits. Your supervisor says you should regard the excess as a mere 'technicality', and he asks you to 'adjust' the data so that the data appears to be in compliance. He says that slight excess is not going to endanger human or fish life any more than if the plant were actually in compliance. However he says, solving the problem would require a very heavy investment. He explains, "We can not afford new equipment's. It might cost even a few jobs. It will set us behind our competitors. Besides, he says that many of the competitors are doing the same and the bad publicity we would get might scare off some of the tourist industry, making it worse for everybody"

Questions:

a) What are your basic responsibilities as an environmental engineer in this plant? How should you respond to your supervisors requests? What are the ethical issues in this case ?

b) Consider the same scenario from the view points of (i) Department of Anti-pollution (ii) Parents of children who swim in the lake (ii) Those who fish in the lake.

c) Should your ideas as environmental engineer change as you take into these different view points? What should you do, from 'all things considered' perspective, as environmental engineer?

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9. LECTURE NOTES
UNIT V GLOBAL ISSUES

1. Multinational Corporations

Organizations who have established business in more than one country, are called multinational corporation. The headquarters are in the home country and the business is extended in many host countries. The Western organizations doing business in the less-economically developed (developing, and overpopulated) countries gain the advantage of inexpensive labor, availability of natural resources, conducive-tax atmosphere, and virgin market for the products. At the same time, the developing countries are also benefited by fresh job opportunities, jobs with higher remuneration and challenges, transfer of technology, and several social benefits by the wealth developed. But this happens invariably with some social and cultural disturbance. Loss of jobs for the home country, and loss or exploitation of natural resources, political instability for the host countries are some of the threats of globalization.

1. International Human Rights

To know what are the moral responsibilities and obligations of the multinational corporations operating in the host countries, let us discuss with the framework of rights ethics. Common minimal rights are to be followed to smoothen the transactions when the engineers and employers of MNCs have to interact at official, social, economic and sometimes political levels. At international level, the organizations are expected to adopt the minimum levels of (a) values, such as mutual support, loyalty, and reciprocity, (b) the negative duty of refraining from harmful actions such as violence and fraud, and (c) basic fairness and practical justice in case of conflicts.

The ten international rights to be taken care of, in this context are:

1. Right of freedom of physical movement of people
2. Right of ownership of properties
3. Freedom from torture

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4. Right to fair trial on the products

5.Freedom from discrimination on the basis of race or sex. If such discrimination against women or minorities is prevalent in the host country, the MNC will be compelled to accept. MNCs may opt to quit that country if the human rights violations are severe.

6.Physical security. Use of safety gadgets have to be supplied to the workers even if the laws of the host country do not suggest such measures.

7. Freedom of speech and forming association

8. Right to have a minimum education

9. Right to political participation

10.Right to live and exist (i.e., coexistence). The individual liberty and sanctity of the human life are to be respected by all societies.

2. Technology Transfer

It is a process of moving technology to a new setting and implementing it there. Technology includes hardware (machines and installations) and the techniques (technical, organizational, and managerial skills and procedures). It may mean moving the technology applications from laboratory to the field/factory or from one country to another. This transfer is effected by governments, organizations, universities, and MNCs.

3. Appropriate Technology

Identification, transfer, and implementation of most suitable technology for a set of new situations, is called appropriate technology. Technology includes both hardware (machines and installations) and software (technical, organizational and managerial skills and procedures). Factors such as economic, social, and engineering constraints are the causes for the modification of technology. Depending on the availability of resources, physical conditions (such as temperature, humidity, salinity, geographical location, isolated land area, and availability of water), capital opportunity costs, and the human value system (social acceptability) which includes their traditions, beliefs, and religion, the appropriateness is to be determined.

For example, small farmers in our country prefer to own and use the power tillers, rather than the high-powered tractors or sophisticated harvesting machines. On the other hand, the latest technological device, the cell phones and wireless local loop phones have found their way into remote villages and hamlets, than the landline telephone connections.

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Large aqua-culture farms should not make the existing fishermen jobless in their own village. The term appropriate is value based and it should ensure fulfillment of the human needs and protection of the environment.

5.1.4 How Appropriate is Aptech?

1.A case against the technology transfer is that the impact of borrowed or transferred technology has been threatening the environment beyond its capacity and sustainable development of the host countries. Large plantations that orient their efforts to exports leave the small farmers out of jobs and at the mercy of the foreign country. For example, genetically modified cotton have shown sufficient disturbance in Europe and Africa. This has made the European Union to oppose the entry of G.M. cotton into Europe.

2.The high technology has contributed to large-scale migration from villages to the cities where corporations are located, leading to the undesirable side-effects of overcrowding of cities, such as the scarcity of water, insanitation, poverty, and the increase in crimes.

3.The term 'appropriate' should emphasize the social acceptability and environmental protection of the host countries, and this need to be addressed while transferring technology. Thus, we confirm the view that engineering is a continual social experimentation with nature.

5.1.5 MNCs and Morality

The economic and environmental conditions of the home and host countries may vary. But the multinational institutions have to adopt appropriate measures not to disturb or dislocate the social and living conditions and cultures of the home countries. A few principles are enlisted here:

1. MNC should respect the basic human rights of the people of the host countries.

2.The activities of the MNC should give economic and transfer technical benefits, and implement welfare measures of the workers of the host countries.

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3. The business practices of the multinational organizations should improve and promote morally justified institutions in the host countries.

4. The multinationals must respect the laws and political set up, besides cultures and promote the cultures of the host countries.

5. The multinational organizations should provide a fair remuneration to the employees of the host countries. If the remuneration is high as that of home country, this may create tensions and if it is too low it will lead to exploitation.

6. Multinational institutions should provide necessary safety for the workers when they are engaged in hazardous activities and 'informed consent' should be obtained from them. Adequate compensation should be paid to them for the additional risks undertaken.

5.1.5 Case Study:

Bhopal Gas Tragedy The Union Carbide had 51% and the Indian subsidiary UC India Ltd. had 49% of stock. In 1983, there were 14 plants in India manufacturing chemicals, pesticides, and other hazardous products. The Bhopal plant had a license to make Methyl isocyanate-based pesticides. In November 1984, they had decided to close down the plant. For quite some years before the production rate was going down. In the history of chemical plants disasters, three other wake-up calls were reported.

Flixborough accident in 1974 in U.K. when certain modifications carried out in the plant led to the leakage and explosion of cyclohexane, which killed 28 people. The Piper Alpha offshore oil platform disaster in 1988, near Scotland, killed 167 people and resulted in $ 2 billion losses. The third occurred in Toulouse, France in 2001, killing 29 people, and injuring thousands. A warehouse holding 300 tones of ammonium nitrate fertilizer exploded and damaged 10000 buildings, including schools, a university, and a hospital. But we have not learnt from the past.

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5.1.6 Ethical Balance

Should an organization adopt the rules and practices of the host country fully and face dangers and other serious consequences or adopt strictly their own country's standards and practices in the host country?

There is a saying, "When in Rome do as Romans do". Can this be applied in the case of MNCs? This is called ethical relativism. The actions of corporation and individuals that are accepted by law, custom and other values of a society can be morally right in that society. It is morally false, if it is illogical. It means, the corporation (and the engineers) functioning in other countries must understand their law, customs, and beliefs and act in line with those prevailing in that country. This will lead to disaster if the country is a developing one where the safety standards are given a go-bye. Laws and conventions are not morally self-sustaining. In a overpopulated country, the loss of human lives may not physically affect them, but the tragedy cast shadow for over decades, as it happened in Bhopal in 1984. This will be criticized from the points of view of human rights, public welfare, and respect to people.

On the other hand, the organizations may practice laws of the home country, without adjustments to the host culture. This stand is called ethical absolutism. This is again false, since the moral principles in a different culture come into conflicts, and implementation in the 'hostile' culture is almost impossible.

Hence, MNCs may adopt ethical relationalism (contexualism) as a compromise. Moral judgments are made in relation to the factors prevailing locally, without framing rigid rules. The judgments should be contextual and in line with the customs of other cultures. The ethical pluralism which views more than one justifiable moral solution is also adaptable. This principle accepts cultural diversity and respects the legitimate cultural differences among individuals and groups, of the host country.

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5.2 Environmental Ethics

Environmental ethics is the study of (a) moral issues concerning the environment, and (b) moral perspectives, beliefs, or attitudes concerning those issues. Engineers in the past are known for their negligence of environment, in their activities. It has become important now that engineers design eco-friendly tools, machines, sustainable products, processes, and projects. These are essential now to (a) ensure protection (safety) of environment (b) prevent the degradation of environment, and (c) slow down the exploitation of the natural resources, so that the future generation can survive. The American Society of Civil Engineers (ASCE) code of ethics, has specifically requires that "engineers shall hold paramount the safety, health, and welfare of the public and shall strive to comply with the principles of sustainable development in the performance of professional duties" The term sustainable development emphasizes on the investment, orientation of technology, development and functioning of organizations to meet the present needs of people and at the same time ensuring the future generations to meet their needs.

Compaq Computer Corporation (now merged with HP) was the leader, who exhibited their commitment to environmental health, through implementation of the concept of 'Design for environment' on their products, unified standards all over the world units, and giving priority to vendors with a record of environmental concern.

Engineers as experimenters have certain duties towards environmental ethics, namely:

1. Environmental impact assessment: One major but sure and unintended effect of technology is wastage and the resulting pollution of land, water, air and even space. Study how the industry and technology affects the environment.
2. Establish standards: Study and to fix the tolerable and actual pollution levels.
3. Counter measures: Study what the protective or eliminating measures are available for immediate implementation
4. Environmental awareness: Study on how to educate the people on environmental practices, issues, and possible remedies..

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5.2.1 Disasters

1. Plastic Waste Disposal

In our country, several crores of plastic bottles are used as containers for water and oil, and plastic bags are used to pack different materials ranging from vegetables to gold ornaments. Hardly any of these are recycled. They end up in gutters, roadsides, and agricultural fields. In all these destinations, they created havoc. The worse still is the burning of plastic materials in streets and camphor along with plastic cover in temples, since they release toxic fumes and threaten seriously the air quality. Cities and local administration have to act on this, collect and arrange for recycling through industries.

2. e-Waste Disposal

The parts of computers and electronic devices which have served its useful life present a major environmental issue for all the developing countries including India. This scrap contains highly toxic elements such as lead, cadmium, and mercury. Even the radioactive waste will lose 89% of its toxicity after 200 years, by which time it will be no more toxic than some natural minerals in the ground. It will lose 99% of its remaining toxicity over the next 30,000 years. The toxic chemical agents such as mercury, arsenic, and cadmium retain toxicity undiminished for ever.

3. Industrial Waste Disposal

There has been a lot of complaints through the media, on (a) against the Sterlite Copper Smelting Plant in Thuthukkudi (1997) against its pollution, and (b) when Indian companies imported the discarded French Warship Clemenceau for disposal, the poisonous asbestos compounds were expected to pollute the atmosphere besides exposing the labor to a great risk, during the disposal. The government did not act immediately. Fortunately for Indians, the French Government intervened and withdrew the ship, and the serious threat was averted!

4. Depletion of Ozone Layer

The ozone layer protects the entire planet from the ill-effects of ultraviolet radiation and is vital for all living organisms in this world. But it is eaten away by the Chloro-fluro-carbons (CFC) such as freon emanating from the refrigerators, air conditioners, and aerosol can spray. This has caused also skin cancer to sun-bathers in the Western countries. Further NO and NO2 gases were also found to react with the ozone. Apart from engineers, the organizations, laws of the country and local administration and market mechanisms are required to take up concerted efforts to protect the environment.

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5. Global Warming

Over the past 30 years, the Earth has warmed by 0.6 °C. Over the last 100 years, it has warmed by 0.8 °C. It is likely to push up temperature by 3°C by 2100, according to NASA's studies. The U.S. administration has accepted the reality of global climate change, which has been associated with stronger hurricanes, severe droughts, intense heat waves and the melting of polar ice. Greenhouse gases, notably carbon dioxide emitted by motor vehicles and coal-fired power plants, trap heat like the glass walls of a greenhouse, cause the Earth to warm up. Delegates from the six countries — Australia, China, India, Japan, South Korea and US met in California in April 2006 for the first working session of the AsiaPacific Partnership on Clean Development and Climate. These six countries account for about half of the world's emissions of climate-heating greenhouse gases. Only one of the six, Japan, is committed to reducing greenhouse gas emissions by at least 5.2 per cent below 1990 levels by 2012 under the Kyoto Agreement.

About 190 nations met in Germany in the middle of May 2006 and tried to bridge vast policy gaps between the United States and its main allies over how to combat climate change amid growing evidence that the world is warming that could wreak havoc by stoking more droughts, heat waves, floods, more powerful storms and raise global sea levels by almost a meter by 2100.

6. Acid Rain

Large emissions of sulphur oxides and nitrous oxides are being released in to the air from the thermal power stations using the fossil fuels, and several processing industries. These gases form compounds with water in the air and precipitates as rain or snow on to the earth. The acid rain in some parts of the world has caused sufficient damage to the fertility of the land and to the human beings.

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5.2.2 Human-centered Environmental Ethics

This approach assumes that only human beings have inherent moral worth duly to be taken care of. Other living being and ecosystems are only instrumental in nature. Utilitarianism aims to maximize good consequences for human beings. Most of the goods are engineered products made out of natural resources. Human beings have also (a) recreational interests (enjoy leisure through mountaineering, sports, and pastimes), (b) aesthetic interests (enjoy nature as from seeing waterfalls and snow-clad mountains), (c) scientific interests to explore into nature or processes, and (d) a basic interest to survive, by preservation as well as conservation of nature and natural resources.

Rights ethicists favor the basic rights to live and right to liberty, to realise the right to a live in a supportive environment. Further, virtue ethics stresses importance of prudence, humility, appreciation of natural beauty, and gratitude to the mother nature that provides everything.

However, the nature-centered ethics, which ensures the worth of all living beings and organisms, seems to be more appropriate in the present-day context. Many Asian religions stress the unity with nature, rather than domination and exploitation. The Zen Buddhism calls for a simple life with compassion towards humans and other animals. Hinduism enshrines the ideal of oneness (advaitha) in and principle of ahimsa to all living beings. It identifies all the human beings, animals, and plants as divine. The eco-balance is the need of the hour and the engineers are the right experimenters to achieve this.

5.3 Computer Ethics

Computer ethics is defined as (a) study and analysis of nature and social impact of computer technology, (b) formulation and justification of policies, for ethical use of computers. This subject has become relevant to the professionals such as designers of computers, programmers, system analysts, system managers, and operators. The use of computers have raised a host of moral concerns such as free speech, privacy, intellectual property right, and physical as well as mental harm. There appears to be no conceptual framework available on ethics, to study and understand and resolve the problems in computer technology.

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1. Types of Issues

Different types of problems are found in computer ethics.

1. Computer as the Instrument of Unethical Acts

(a)The usage of computer replaces the job positions. This has been overcome to a large extent by readjusting work assignments, and training everyone on computer applications such as word processing, editing, and graphics.

(b)Breaking privacy. Information or data of the individuals accessed or erased or the ownership changed.

(c)Defraud a bank or a client, by accessing and withdrawing money from other's bank account.

2. Computer as the Object of Unethical Act The data are accessed and deleted or changed.

(a)Hacking: The software is stolen or information is accessed from other computers. This may cause financial loss to the business or violation of privacy rights of the individuals or business. In case of defense information being hacked, this may endanger the security of the nation.

(b) **Spreading virus:** Through mail or otherwise, other computers are accessed and the files are erased or contents changed altogether. 'Trojan horses' are implanted to distort the messages and files beyond recovery. This again causes financial loss or mental torture to the individuals. Some hackers feel that they have justified their right of free information or they do it for fun. However, these acts are certainly unethical.

(c)Health hazard: The computers pose threat during their use as well as during disposal.

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3. Problems Related to the Autonomous Nature of Computer

(a) Security risk: Recently the Tokyo Stock Exchange faced a major embarrassment. A seemingly casual mistake by a junior trader of a large security house led to huge losses including that of reputation. The order through the exchange's trading system was to sell one share for 600,000 Yen. Instead the trader keyed in a sale order for 600,000 shares at the rate of one Yen each. Naturally the shares on offer at the ridiculously low price were lapped up. And only a few buyers agreed to reverse the deal! The loss to the securities firm was said to be huge, running into several hundred thousands. More important to note, such an obvious mistake could not be corrected by some of the advanced technology available. For advanced countries like Japan who have imbibed the latest technology, this would be a new kind of learning experience.12

(b) Loss of human lives: Risk and loss of human lives lost by computer, in the operational control of military weapons. There is a dangerous instability in automated defense system. An unexpected error in the software or hardware or a conflict during interfacing between the two, may trigger a serious attack and cause irreparable human loss before the error is traced. The Chinese embassy was bombed by U.S. military in Iraq a few years back, but enquiries revealed that the building was shown in a previous map as the building where insurgents stayed.

(c) In flexible manufacturing systems, the autonomous computer is beneficial in obtaining continuous monitoring and automatic control. Various issues related to computer ethics are discussed as follows:

5.3.2 Computers In Workplace

The ethical problems initiated by computers in the workplace are:

1**. Elimination of routine and manual jobs.** This leads to unemployment, but the creation of skilled and IT-enabled service jobs are more advantageous for the people.

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Initially this may require some upgradation of their skills and knowledge, but a formal training will set this problem right. For example, in place of a typist, we have a programmer or an accountant.

2.Health and safety: The ill-effects due to electromagnetic radiation, especially on women and pregnant employees, mental stress, wrist problem known as Carpel Tunnel Syndrome, and backpain due to poor ergonomic seating designs, and eye strain due to poor lighting and flickers in the display and long exposure, have been reported worldwide. Over a period of long exposure, these are expected to affect the health and safety of the people. The computer designers should take care of these aspects and management should monitor the health and safety of the computer personnel.

3.Computer failure: Failure in computers may be due to errors in the hardware or software. Hardware errors are rare and they can be solved easily and quickly. But software errors are very serious as they can stop the entire network. Testing and quality systems for software have gained relevance and importance in the recent past, to avoid or minimize these errors.

5.3.3 Property Issues

The property issues concerned with the computers are:

1. Computers have been used to extort money through anonymous telephone calls.
2. Computers are used to cheat and steal by current as well as previous employees.
3. Cheating of and stealing from the customers and clients.
4. Violation of contracts on computer sales and services.
5. Conspiracy as a group, especially with the internet, to defraud the gullible, stealing the identity and to forge documents.
6. Violation of property rights: Is the software a property? The software could be either a Program (an algorithm, indicating the steps in solving a problem) or a Source code (the algorithm in a general computer language such as FORTAN, C and COBOL or an Object code (to translate the source code into the machine language). How do we apply the concept of property here? This demands a framework for ethical judgments.

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Property is what the laws permits and defines as can be owned, exchanged, and used. The computer hardware (product) is protected by patents. The software (idea, expression) is protected by copyrights and trade secrets. But algorithms can not be copyrighted, because the mathematical formulas can be discovered but not owned. The object codes which are not intelligible to human beings can not be copyrighted.

Thus, we see that reproducing multiple copies from one copy of (licensed) software and distribution or sales are crimes. The open source concepts have, to a great extent, liberalized and promoted the use of computer programs for the betterment of society.

5.3.4 Computer Crime

The ethical features involved in computer crime are:

1.Physical Security The computers are to be protected against theft, fire, and physical damage. This can be achieved by proper insurance on the assets.

2.Logical security The aspects related are (a) the privacy of the individuals or organizations, (b) confidentiality, (c) integrity, to ensure that the modification of data or program are done only by the authorized persons, (d) uninterrupted service. This is achieved by installing appropriate uninterrupted power supply or back-up provisions, and (e) protection against hacking that causes dislocation or distortion. Licensed anti-virus packages and firewalls are used by all computer users to ensure this protection. Passwords and data encryption have been incorporated in the computer software as security measures. But these have also been attacked and bye-passed. But this problem is not been solved completely.

Major weaknesses in this direction are: (a) the difficulty in tracing the evidence involved and (b) absence of stringent punishment against the crime. The origin of a threat to the Central Government posted from an obscure browsing center, remained unsolved for quite a long time. Many times, such crimes have been traced, but there are no clear cyber laws to punish and deter the criminals.

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5.3.5 Privacy and Anonymity

The data transmission and accessibility have improved tremendously by using the computers, but the right to privacy has been threatened to a great extent. Some issues concerned with the privacy are listed hereunder:

1. **Records of Evidence:** Service records or criminal records and the details of people can be stored and accessed to prove the innocence or guilty. Records on psychiatric treatment by medical practitioners or hospital, or records of membership of organizations may sometime embarrass the persons in later years.

2. **Hacking:** There are computer enthusiasts who willfully or for fun, plant virus or "Trojan horses" that may fill the disc space, falsify information, erase files, and even harm the hardware. They breakdown the functioning of computers and can be treated as violation of property rights. Some hackers opine that the information should be freely available for everybody. It is prudent that the right to individual privacy in limiting the access to the information on oneself, should not be violated. Further any unauthorized use of personal information (which is a property), is to be considered as theft. Besides the individual privacy, the national security, and freedom within the economy are to be respected. The proprietary information and data of the organizations are to be protected so that they can pursue the goals without hindrance.

3. **Legal Response:** In the Indian scene, the Right to Information Act 2005 14 provides the right to the citizens to secure access to information under the control of public authorities, including the departments of the central government, state governments, government bodies, public sector companies and public sector banks, to promote transparency and accountability of public authorities. Right to information: Under the Act, section 2 (j), the right to information includes the right to (1) Inspect works, documents, records, (2) take notes, extracts or certified copies of documents or records, (3) take certified samples of material, and (4) obtain information in the form of printouts, diskettes, floppies, tapes, video cassettes or in any other electronic mode.

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4. Anonymity: Anonymity in the computer communication has some merits as well as demerits. While seeking medical or psychological counseling or discussion (chat) on topics, such as AIDS, abortion, gay rights, the anonymity offers protection (against revealing their identity). But frequently, anonymity is misused by some people for money laundering, drug trafficking and preying upon the vulnerable.

6. Professional Responsibility

The computer professionals should be aware of different conflicts of interests as they transact with other at different levels. The IEEE and Association for Computing Machinery (ACM) have established the codes of ethics to manage such responsibilities.

7. The Big Net

Almost all the countries are now connected by the internet. But there are no international laws to regulate the issues of freedom of speech, intellectual property rights, privacy rights etc. Another development in this direction is, the universities offering degrees-on-line. Third World is certainly gaining knowledge and education. Even Google.com has announced plans to publish research papers through the World Wide Web. Knowledge is power. Knowledge is internationalised! Will this lead to empowerment of the Third World and promotion of World peace? Only the future can answer this question.

5.4 Weapons Development

Military activities including the world wars have stimulated the growth of technology. The growth of Internet amply illustrates this fact. The development of warfare and the involvement of engineers bring out many ethical issues concerned with engineers, such as the issue of integrity in experiments as well as expenditure in defense research and development, issue of personal commitment and conscience, and the issues of social justice and social health.

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Engineers involve in weapons development because of the following reasons:

1. It gives one job with high salary.
2. One takes pride and honor in participating in the activities towards the defense of the nation (patriotic fervor).
3. One believes the he fights a war on terrorism and thereby contribute to peace and stability of the country. Ironically, the wars have never won peace, only peace can win peace!
4. By research and development, the engineer is reducing or eliminating the risk from enemy weapons, and saving one's country from disaster.
5. By building-up arsenals and show of force, a country can force the rogue country, towards regulation. Engineers can participate effectively in arms control negotiations for surrender or peace, e.g., bombing of Nagasaki and Hiroshima led to surrender by the Japanese in 1945. Many engineers had to fight and convince their personal conscience. The scene such as that of a Vietnamese village girl running wild with burns on the body and horror in the face and curse in her mind has moved some enginee

5. Engineers As Managers

1. Characteristics

The characteristics of engineers as managers are:

1. Promote an ethical climate, through framing organization policies, responsibilities and by personal attitudes and obligations.
2. Resolving conflicts, by evolving priority, developing mutual understanding, generating various alternative solutions to problems.
3. Social responsibility to stakeholders, customers and employers. They act to develop wealth as well as the welfare of the society.

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Ethicists project the view that the manager's responsibility is only to increase the profit of the organization, and only the engineers have the responsibility to protect the safety, health, and welfare of the public. But managers have the ethical responsibility to produce safe and good products (or useful service), while showing respect for the human beings who include the employees, customers and the public. Hence, the objective for the managers and engineers is to produce valuable products that are also profitable.

5.5.2 Managing Conflicts

In solving conflicts, force should not be resorted. In fact, the conflict situations should be tolerated, understood, and resolved by participation by all the concerned. The conflicts in case of project managers arise in the following manners:

(a) Conflicts based on schedules: This happens because of various levels of execution, priority and limitations of each level.

(b) Conflicts arising out of fixing the priority to different projects or departments. This is to be arrived at from the end requirements and it may change from time to time.

(c) Conflict based on the availability of personnel.

(d) Conflict over technical, economic, and time factors such as cost, time, and performance level.

(e) Conflict arising in administration such as authority, responsibility, accountability, and logistics required.

(f) Conflicts of personality, human psychology and ego problems.

(g) Conflict over expenditure and its deviations.

Most of the conflicts can be resolved by following the principles listed here:

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1. People

Separate people from the problem. It implies that the views of all concerned should be obtained. The questions such as what, why, and when the error was committed is more important than to know who committed it. This impersonal approach will lead to not only early solution but also others will be prevented from committing errors.

2. Interests

Focus must be only on interest i.e., the ethical attitudes or motives and not on the positions (i.e., stated views). A supplier may require commission larger than usual prevailing rate for an agricultural product. But the past analysis may tell us that the material is not cultivated regularly and the monsoon poses some additional risk towards the supply. Mutual interests must be respected to a maximum level. What is right is more important than who is right!

3. Options

Generate various options as solutions to the problem. This helps a manager to try the next best solution should the first one fails. Decision on alternate solutions can be taken more easily and without loss of time.

4. Evaluation

The evaluation of the results should be based on some specified objectives such as efficiency, quality, and customer satisfaction. More important is that the means, not only the goals, should be ethical.

5.6 Consulting Engineers

The consulting engineers work in private. There is no salary from the employers. But they charge fees from the sponsor and they have more freedom to decide on their projects. Still they have no absolute freedom, because they need to earn for their living. The consulting engineers have ethical responsibilities different from the salaried engineers, as follows:

1. Advertising The consulting engineers are directly responsible for advertising their services, even if they employ other consultants to assist them. But in many organisations, this responsibility is with the advertising executives and the personnel department.

They are allowed to advertise but to avoid deceptive ones. Deceptive advertising such as the following are prohibited:

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(a) By white lies.

(b)Half-truth, e.g., a product has actually been tested as prototype, but it was claimed to have been already introduced in the market. An architect shows the photograph of the completed building with flowering trees around but actually the foundation of the building has been completed and there is no real garden.

(c)Exaggerated claims. The consultant might have played a small role in a well-known project. But they could claim to have played a major role.

(d)Making false suggestions. The reduction in cost might have been achieved along with the reduction in strength, but the strength details are hidden.

(e) Through vague wordings or slogans.

2. Competitive Bidding

It means offering a price, and get something in return for the service offered. The organizations have a pool of engineers. The expertise can be shared and the bidding is made more realistic. But the individual consultants have to develop creative designs and build their reputation steadily and carefully, over a period of time. The clients will have to choose between the reputed organizations and proven qualifications of the company and the expertise of the consultants. Although competent, the younger consultants are thus slightly at a disadvantage.

3. Contingency Fee

This is the fee or commission paid to the consultant, when one is successful in saving the expenses for the client. A sense of honesty and fairness is required in fixing this fee. The NSPE Code III 6 (a) says that the engineers shall not propose or accept a commission on a contingent basis where their judgment may be compromised.

The fee may be either as an agreed amount or a fixed percentage of the savings realized. But in the contingency fee-agreements, the judgment of the consultant may be biased. The consultant may be tempted to specify inferior materials or design methods to cut the construction cost. This fee may motivate the consultants to effect saving in the costs to the clients, through reasonably moral and technological means

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4. Safety and Client's Needs

The greater freedom for the consulting engineers in decision making on safety aspects, and difficulties concerning truthfulness are the matters to be given attention. For example, in design-only projects, the consulting engineers may design something and have no role in the construction. Sometimes, difficulties may crop-up during construction due to non-availability of suitable materials, some shortcuts in construction, and lack of necessary and adequate supervision and inspection. Properly-trained supervision is needed, but may not happen, unless it is provided. Further, the contractor may not understand and/or be willing to modify the original design to serve the clients best.

A few on-site inspections by the consulting engineers will expose the deficiency in execution and save the workers, the public, and the environment that may be exposed to risk upon completion of the project.

The NSPE codes on the advertisement by consultants provide some specific regulations. The following are the activities prohibited in advertisement by consultant:

1. The use of statement containing misrepresentation or omission of a necessary fact.
2. Statement intended or likely to create an unjustified expectation.
3. Statement containing prediction of future (probable) success.
4. Statement intended or likely to attract clients, by the use of slogans or sensational language format.

5.7 Engineers As Expert Witness

Frequently engineers are required to act as consultants and provide expert opinion and views in many legal cases of the past events. They are required to explain the causes of accidents, malfunctions and other technological behavior of structures, machines, and instruments, e.g., personal injury while using an instrument, defective product, traffic accident, structure or building collapse, and damage to the property, are some of the cases where testimonies are needed. The focus is on the past.

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The functions of eye-witness and expert-witness are different as presented in the Table 5.1.

Eye-witness	Expert-witness
1. Eye-witness gives evidence on only what has been seen or heard actually	1.Gives expert view on the facts in their area of their expertise
	2.Interprets the facts, in term of the cause and effect relationship
	3.Comments on the view of the opposite side
	4. Reports on the professional standards, especially on the precautions when the product is made or the service is provided

Table 5.1 Eye-witness and expert-witness

The engineers, who act as expert-witnesses, are likely to abuse their positions in the following manners:

1.Hired Guns Mostly lawyers hire engineers to serve the interest of their clients. Lawyers are permitted and required to project the case in a way favorable to their clients. But the engineers have obligations to thoroughly examine the events and demonstrate their professional integrity to testify only the truth in the court. They do not serve the clients of the lawyers directly. The hired guns forward white lies and distortions, as demanded by the lawyers. They even withhold the information or shade the fact, to favor their clients.

2.Money Bias Consultants may be influenced or prejudiced for monitory considerations, gain reputation and make a fortune.

3.Ego Bias The assumption that the own side is innocent and the other side is guilty, is responsible for this behavior. An inordinate desire to serve one's client and get name and fame is another reason for this bias.

4.Sympathy Bias Sympathy for the victim on the opposite side may upset the testimony. The integrity of the consultants will keep these biases away from the justice. The court also must obtain the balanced view of both sides, by examining the expert witnesses of lawyers on both sides,3t8o remove a probable bias.

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5.7.1 Duties

1. The expert-witness is required to exhibit the responsibility of confidentiality just as they do in the consulting roles. They can not divulge the findings of the investigation to the opposite side, unless it is required by the court of law.

2. More important is that as witness they are not required to volunteer evidence favorable to the opponent. They must answer questions truthfully, need not elaborate, and remain neutral until the details are asked for further.

3. They should be objective to discover the truth and communicate them honestly.

4. The stand of the experts depends on the shared understanding created within the society. The legal system should be respected and at the same time, they should act in conformance with the professional standards as obtained from the code of ethics.

5. The experts should earnestly be impartial in identifying and interpreting the observed data, recorded data, and the industrial standards. They should not distort the truth, even under pressure. Although they are hired by the lawyers, they do not serve the lawyers or their clients. They serve the justice. Many a time, their objective judgments will help the lawyer to put up the best defense for their clients.

5.8 Moral Leadership

Engineers provide many types of leadership in the development and implementation of technology, as managers, entrepreneurs, consultants, academics and officials of the government. Moral leadership is not merely the dominance by a group. It means adopting reasonable means to motivate the groups to achieve morally desirable goals. This leadership presents the engineers with many challenges to their moral principles.

Moral leadership is essentially required for the engineers, for the reasons listed as follows:

1. It is leading a group of people towards the achievement of global and objectives. The goals as well as the means are to be moral. For example, Hitler and Stalin were leaders, but only in an instrumental sense and certainly not on moral sense.

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2.The leadership shall direct and motivate the group to move through morally desirable ways.

3.They lead by thinking ahead in time, and morally creative towards new applications, extension and putting values into practice. 'Morally creative' means the identification of the most important values as applicable to the situation, bringing clarity within the groups through proper communication, and putting those values into practice.

4.They sustain professional interest, among social diversity and cross-disciplinary complexity. They contribute to the professional societies, their professions, and to their communities. The moral leadership in engineering is manifested in leadership within the professional societies. The professional societies provide a forum for communication, and canvassing for change within and by groups.

5.Voluntarism: Another important avenue for providing moral leadership within communities, by the engineers is to promote services without fee or at reduced fees (pro bono) to the needy groups. The professional societies can also promote such activities among the engineers. This type of voluntarism (or philanthropy) has been in practice in the fields of medicine, law and education. But many of the engineers are not self-employed as in the case of physicians and lawyers.

6.Community service: This is another platform for the engineers to exhibit their moral leadership. The engineers can help in guiding, organising, and stimulating the community towards morally- and environmentally-desirable goals. The corporate organizations have come forward to adopt villages and execute many social welfare schemes, towards this objective.

The Codes of Ethics promote and sustain the ethical environment and assist in achieving the ethical goals in the following manner:

1. It creates an environment in a profession, where ethical behavior is the basic criterion.
2. It guides and reminds the person as to how to act, in any given situation. 3.It provides support to the individual, who is being pressurized or tortured by a superior or employer, to behave unethically.

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It provides support to the individual, who is being pressurized or tortured by a superior or employer, to behave unethically.

4. Apart from professional societies, companies and universities have framed their own codes of ethics, based on the individual circumstances and specific mission of the organisations. These codes of conduct help in employees' awareness of ethical issues, establish, and nurture a strong corporate ethical culture.

9. –Code of Conduct

1.National Society of Professional Engineers Preamble Engineering is an important and learned profession. As members of this profession, engineers are expected to exhibit the higher standards of honesty and integrity. Engineering has a direct and vital impact on the quality of life for all people. Accordingly, the services provided by engineers require honesty, impartiality, fairness, and equity, and must be dedicated to the protection of then public health, safety, and welfare. Engineer must perform under a standard of professional behavior that requires adherence to the highest principles of ethical conduct.

I Fundamental Canons

Engineers in the fulfillment of their professional duties shall

1. hold paramount the safety, health, and welfare of the public.
2. perform services only in areas of their competence.
3. issue public statements only in objective and truthful manner.
4. act for each employer or client as faithful agents or trustees.

5. avoid deceptive acts.

6.conduct themselves honorably, responsibly, ethically, and lawfully so as to enhance the honor, reputation, and usefulness of the profession.

II Rules of Practice

1. Engineers shall hold paramount the safety, health, and welfare of the public.

(a) If engineers' judgment is overruled under circumstances that endanger life or property, they shall notify their employer or client and such other authority as may be appropriate.

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(b)Engineers shall approve only those engineering documents that are in conformity with applicable standards.

(c)Engineers shall not reveal facts, data, or information without prior consent of the client or employer except as authorized or required by law or this code.

(d)Engineers shall not permit the use of their name or associate in business ventures with any person or firm that they believe are engaged in fraudulent or dishonest enterprise.

(e)Engineers shall not aid or abet the unlawful practice of engineering by a person or firm. (f) Engineers having knowledge of any alleged violation of this Code shall report thereon to appropriate professional bodies and when relevant, also to public authorities, and cooperate with the proper authorities in furnishing such information or assistance as may be required. 2. Engineers shall perform services only in the areas of their competence.

(a) Engineers shall undertake assignments only when qualified by education or experience in the specific technical fields involved.

(b) Engineers shall not affix their signatures to any plans or documents dealing with the subject matter in which they lack competence, nor to any plan or document not prepared under their direction and control.

(c) Engineers may accept assignments and assume responsibility for coordination of an entire project and sign and seal the engineering documents for the entire project, provided that each technical segment is signed and sealed only by the qualified engineers who prepared the segment.

3.Engineers shall issue public statements only in an objective and truthful manner.

(a) Engineers shall be objective and truthful in professional reports, statements, or testimony. They shall include all relevant and pertinent information in such reports, statements, or testimony, which should bear the date indicating when it was current.

(b) Engineers may express publicly technical options that are founded upon knowledge of the facts and competence in the subject matter.

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Engineers shall issue no statements, criticisms, or arguments on technical matters that are inspired or paid for by interested parties on prefaced their comments by explicitly identifying the interested parties on whose behalf they are speaking and by revealing the existence of any interest the engineers may have in the matters.

4. Engineers shall at for each employer or client as faithful agents or trustees

(a) Engineers shall disclose all known or potential conflicts of interest that could influence or appear to influence their judgment or the quality of their services.

(b) Engineers shall not accept compensation, financial or otherwise, from more than one party for services on the same project, or for services pertaining to the same project, unless the circumstances are fully disclosed and agreed to by all interested parties.

(c) Engineers shall not solicit or accept financial or other valuable consideration, directly or indirectly, from outside agents on connection with the work for which they are responsible.

(d) Engineers in public service as members, advisers, or employees of a governmental or quasi-governmental body or department shall not participate in decisions with respect to services solicited or provided by them or their organizations in private or public engineering practice.

(e) Engineers shall not solicit or accept a contract from a governmental body on which a principal or officer of their organization serves as a member.

5.Engineers shall avoid deceptive acts

(a) Engineers shall not falsify their qualifications or permit misrepresentation of their or their associate's qualifications. They shall not misrepresent or exaggerate their responsibility in or for the subject matter of prior assignments. Brochures or other presentations incident to the solicitation of employment shall not misrepresent pertinent facts concerning employers, employees, associates, joint ventures, or past accomplishments.

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(b) Engineers shall not offer, give, solicit or receive, either directly or indirectly, any contribution to influence the award of a contract by public authority, or which may be reasonably construed by the public as having the effect of intent to influence the awarding of a contract. They shall not offer any gift or other valuable consideration in order to secure work. They shall not pay a commission, percentage, or brokerage fee in order to secure work, except to a bonafide employee or established commercial or marketing agencies retained by them.

5.9.2 The Institute of Electrical & Electronics Engineers Code of Ethics We the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

1. to accept responsibility in making engineering decisions consistent with the safety, health and welfare of the public, and to disclose prompt factors that might endanger the public or the environment.
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist.
3. to be honest and realistic in stating claims or estimates based on available data.
4. to reject bribery in all its forms.
5. to improve the understanding of technology, its appropriate application, and potential consequences.
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations.
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others.
8. to treat fairly all persons regardless of such factors as race, religion, gender, disability, age, or national origin.
9. to avoid injuring others, their property, reputation, or employment by false or malicious action.
10. to assist colleagues and co-workers in their professional development and to support them in following code of ethics.

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5.9.3 Indian Institute of Material Management

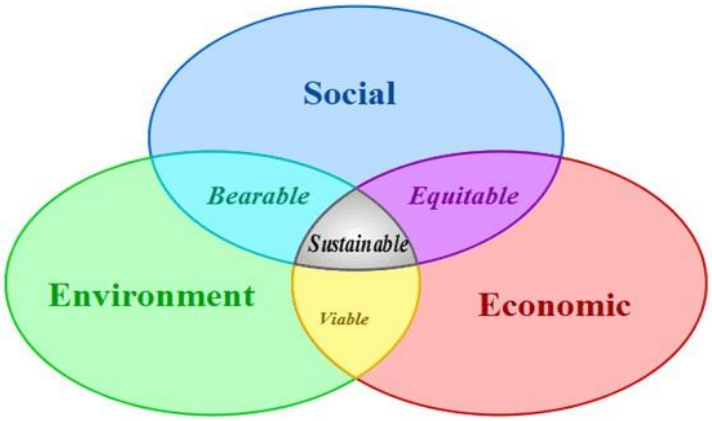
Code of Ethics

1. To consider first the total interest of one's organization in all transactions without impairing the dignity and responsibility to one's office.
2. To buy without prejudice seeking to obtain the maximum ultimate value for each rupee of expenditure.
3. To subscribe and work for honesty and truth in buying and selling.
4. To denounce all forms and manifestations of commercial bribery and to eschew anti-social practices.
5. To respect one's obligations and those of one's organization consistent with good business practice.

5.10 Corporate Social Responsibility

Corporate social responsibility means an organizations have also ethical and societal responsibilities that go beyond their economic responsibilities.

Corporate social responsibility (CSR) is a type of business self-regulation with the aim of being socially accountable. There is no one "right" way companies can practice CSR; many corporate CSR initiatives strive to positively contribute to the public, the economy or the environment. In today's socially conscious environment, employees and customers place a premium on working for and spending their money with businesses that prioritize CSR.



Video Links
UNIT V GLOBAL ISSUES

S.No	Topic	Video Link
1	Corporate Social Responsibility	https://www.youtube.com/watch?v=2fIrfyxe2IU
2	Moral Leadership	https://www.youtube.com/watch?v=Nmsl1xVx7Ec
3	Environmental Ethics	https://www.youtube.com/watch?v=pCqwaUJv4pg
4	Multinational Corporation	https://www.youtube.com/watch?v=OGg9K1wymWk

10. Assignment Unit V

1) Explain in detail the issues pertaining to environment issues? (CO5)

2) Describe the Bhopal Gas Tragedy and its effects? (CO5)

3) Discuss the following in detail

i. Engineers as Managers (CO6)

ii. Engineers as advisors (CO6)

iii. Engineers as consultant (CO6)

iv. Moral leadership (CO6)

4) Describe the destructive nature and power of weapons and their development? (CO5)

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11. Part A Question & Answer

1. What is embezzlement? (CO5) (K1)

The process of computing computer crimes such as stealing or cheating clients and conspiracy in the fraudulent uses of computer networks is called embezzlement.

2. What the hired guns? (CO6) (K1)

Engineers are hired by attorneys to help them to establish the facts in away favorable to their clients. The hired guns violate the standards of honesty and also due care in conducting investigations.

3. What is technology transfer? (CO6) (K1)

Technology transfer is a process of changing the technology to a new setting and implementing it. Technology includes hardware such as machines and installations as well as techniques such as technical, organizational and managerial skills and procedures.

4. What does moral leadership mean? (CO6) (K1)

Whenever the goals of a leader become permissible and also morally valuable, it is known as moral leadership. Moral leadership also means that employing morally acceptable ways to motivate the groups to move towards morally desirable ways. The ways are depending on the situations.

5.State the most important ethical mistake made by the multinational corporation which caused Bhopal gas plant disaster. (CO5) (K1)

The tanks used to store Methyl Iso-cyanate were overloaded to a tune of 75%.

- The emergency plant was also filled with a large amount of chemicals.
- The entire refrigeration unit had been shut down as a measure to reduce the cost and this led to increase of temperatures to a higher level.
- One of the disappointed workers unscrewed a pressure gauge on a tank and inserted a hosepipe into it, knowing that it would cause damage, but not to this extent.
- Scrubber has also been shut down.
- Flare tower was also not in an operating condition.

6. Define Conflict resolution. (CO5) (K1)

Conflict resolution is the result based on some objective standard and corporate usually uses general standards for evaluating the results.

7. What is contextualize? (CO5) (K1)

In accordance to Gilligan women try hard to preserve personal relationship will all people. This context-oriented emphasis on maintaining personal relationship is called as ethics of care in contrast with ethics of rules and rights.

8. What are ethical pluralism and ethical relativism? (CO5) (K1)

Ethical pluralism: According to this view there may be alternative moral perspectives that are reasonable, but no one of which must be accepted completely by all rational and morally concerned persons.

Ethical relativism: Actions are morally right when they are approved by law or custom they are wrong when they violate laws or customers.

9.What should an ethical expert witness, even though hired by a company, expected to do? (CO6) (K1)

Engineers should not become the hired-guns to their clients, but instead remain as objective as humanly possible in their investigations and the conclusions they reach .they should avoid biases resulting from money ego, and sympathy.

10. What are the international rights listed by Donaldson? (CO5) (K1)

Thomas Donaldson in his book _The ethics of International Business„ has listed the following as the International rights:

- The right to freedom of physical movement.
- The right to ownership of property .
- The right to freedom from torture.
- The right to a fair trial.

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- The right to nondiscriminatory treatment.
- The right to physical security.
- The right to freedom of speech and association .
- The right to minimal education.
- The right to political participation.
- The right to subsistence.

11. Define appropriate technology? (CO5) (K1)

Appropriate technology refers to the identification, transfer and implementation of the most suitable technology for a new set of conditions.

12. List out four examples for Multinational Corporation. (CO5) (K1)

Large corporations having investment and business in number of countries are known as Multinational or Transnational corporation. Some of them are : Hindustan Lever, Ford, Toyota, Sony, LG, Smith Kline Beecham, ITC, Ponds etc.

13. Define computer ethics? (CO5)(K1)

Computers contribute to a variety of moral problems. In order to evaluate andact appropriately with such problems, a new field of applied ethics termed as _computer ethics„ has been developed. The study of ethical issues that are associated with computer, its peripheral and accesses series and the computing profession is called as computer ethics.

14. What is meant by globalization? (CO5) (K1)

Our lives are increasingly dependent upon the goods/services produced over the world and are influenced by the business from around all the corners of the world. In general world has become a global village and have a global economy. The increasing international flow of capital, technology, trade, and people have had the effects of changing the nature of local organizations governments and people of countries and have led to social changes and developments.

15. What are the three senses of relative values? (CO5) (K1)

Ethical Relativism.

- Descriptive Relativism.
- Moral Relativism.

16.What are the normal issues arise in Multinational Corporation? (CO5) (K1)

Ethical dilemmas faced by certain companies may be specific to their industry or company, other types of ethical issues are common to all types of companies. Handling ethical decisions with wisdom is especially important for small businesses, given the potentially devastating effects these companies may face if such issues aren't handled correctly.

17.Differentiate the Eye witness and expert witness in the legal system? (CO6) (K1)

An eyewitness is one who testifies what they perceived through his or her senses (e.g. seeing, hearing, smelling, touching). That perception might be either with the unaided human sense or with the aid of an instrument, e.g., microscope or stethoscope, or by other scientific means, e.g.a chemical reagent which changes color in the presence of a particular substance

An expert witness is one who allegedly has specialized knowledge relevant to the matter of interest, which knowledge purportedly helps to either make sense of other evidence, including other testimony, documentary evidence or physical evidence (e.g., a fingerprint)

18. What is meant by Moral Leadership? (CO6) (K1)

Moral Leadership is a very different kind of leadership. Rather than aspiring to being followed, **Moral Leaders** aim to serve. Instead of showcasing their own skills, Moral Leaders tend to develop the capacities of others.

19. Define the term "Honesty" and "Moral leadership" (CO6) (K1)

Honesty :A facet of moral character that connotes positive and virtuous attributes such as integrity, truthfulness, and straightforwardness, along with the absence of lying, cheating, or theft

MoralL eadership: A process of social influence in which one person enlists the aid and support of others in accomplishing a common task.

20. What do you understand by "business ethics"? (CO5) (K1)

Business ethics (also corporate ethics) is a form of applied ethics or professional ethics that examines ethical principles and moral or ethical problems that arise in a business environment. It applies to all aspects of business conduct and is relevant to the conduct of individuals and entire organizations.

21. What is the nature of work for Engineers as Consulting Engineers? (CO6) (K1)

- Consulting engineers work in private practice.
- They earn by getting their fee for services rendered.
- They have greater freedom in decision making compared to salaried employees.
- But they also have a need to earn a living.

22. What are the major areas of work for engineer consultants? (CO6) (K1)

1. Advertising
2. Competitive bidding
3. Contingency fees
4. Safety and client needs

23. When does the 'contingency fee' become permissible? (CO5) (K1)

To decide whether "contingency fee" practice may be allowed or not, the potential gains should be weighed against the potential losses. Hence, this again calls for contextual reasoning based on ethical theories, which provide a framework for assessing morally relevant issues of the problem.

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24. What are the different views on 'leadership in communities'? (CO6) (K1)

• One view is that no one is strictly obligated to participate in public decision making. It may be a moral ideal for citizens.

• An opposite view is that all are obligated to devote sometime and energy in public policy making.

•Non-engineers should at least stay informed about public issues and professionals have obligations as experts in their areas.

• Hence the need for identifying and expanding areas of possible good.

25. What should engineers do in taking part in Weapons development? (CO6) (K1)

Engineers need to examine one"s conscience to take part in any form of weapon development.

oThey have to consider the circumstances leading to the specific conflict and decide whether it is justified to take part in associated weapons development.

o If necessary, they should refuse to be a part of it and be prepared to face consequences.

26. When is advertising considered to be deceptive? (CO6)(K1)

Example 1: A consulting firm played actually a very minor role in a well-known project

Situation 1: Its brochure claims that it played a major role.

Situation 2: It makes no claim but only shows the picture of the project.

Situation 3: It shows the picture along with a footnote in fine print the true details about its role in the project.

Situation 4: If the same statement is printed in larger type and not as footnote.

27. What are the senses of relative values? CO6 (K1)

Ethical relativism, Descriptive relativism and moral relativism (or contextualism)

28. How do you contrast ethical relativism and descriptive relativism? CO6 (K1)

Ethical relativism says that actions are morally right when they are approved by law or custom; they are wrong when they violate laws and customs. Descriptive relativism states that beliefs about values differ from culture to culture.

29. What is meant by contextualism? (CO6)(K1)

Contextualism views that moral judgement should be made in relation to factors that may vary from case to case, because formulating simple and absolute rules involving moral judgement is impossible. In other words moral judgements are contextual in nature.

30. What is ecocentric ethics? (CO6)(K1)

It emphasizes inherent value in ecological systems.

31. What are human-centered environmental ethics? CO6 (K1)

Human centered environmental ethics extends traditional ethical theories in the aspect of the threats to human beings presented by the destruction of nature.

32. What are the different categories of problems that exist in computer ethics? CO6 (K1)

Those ethical problems for which the computer is the instrument of the unethical act. For example, the use of a computer to defraus the bank.

Those problems for which the computer is the object of the unethical act. For example, stealing computer software and installing it on one's own computer to access others' information.

Those problems associated with the autonomous nature of computers.

33. What is embezzlement? CO6 (k1)

The process of committing computer crimes such as stealing or cheating clients or customers and conspiracy in the fraudulent uses of computer networks is called embezzlement.

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12. PART B QUESTIONS

1. Explain in detail the issues pertaining to environment issues? **(CO5) (K1)**

2. Describe the Bhopal Gas Tragedy and its effects? **(CO5) (K1)**
3. Discuss the following in detail:

    Business Ethics **(CO5) (K1)**
    Environmental Ethics **(CO5) (K1)**
    Computer Ethics **(CO5) (K1)**
    Weapons Development. **(CO5) (K1)**

4.What is environment ethics?Why it is important to study. Discuss any environment issues in the ethical point of view to engineers. **(CO5) (K1)**

4. Discuss the following in detail:
a) Engineers as Managers **(CO6) (K1)**
b) Engineers as advisors **(CO6) (K1)**
c) Engineers as consultant **(CO6) (K1)**
d) Moral leadership**(CO6) (K1)**

6. Discuss the various global issues that have an impact on business.**(CO5) (K1)**

7. Discuss the ethical issues related to computer ethics and internet. **(CO5) (K1)**

8.Discuss in detail about moral and ethical issues involved in use of Computer Technology. **(CO5) (K1)**

9.State an illustrative case study that touches upon some fundamental issues in environmental ethics.**(CO5) (K1)**

10.Discuss the ethical issues related to computer and internet. **(CO5) (K1)**

11.Do engineers have a moral right to carry out what they consider to be unethical activity? Explain in detail with a case study. **(CO5) (K1)**

12. What do environmental ethics deal with? Discuss the holistic approach of environmental ethics. Write a note on acid rain. **(CO5) (K1)**

13.What is the basic ethical and moral responsibility of a manager-engineer? **(CO5) (K1)**

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13. Supportive Online Certification Courses
UNIT 5

S.No.	Name of the Course	Certification Platform	Website link
1.	Environmental Management & Ethics	Coursera Technical University of Denmark	https://www.coursera.org/lecture/environmental-management-ethics/introduction-to-environmental-ethics-0b14B
2.	Ethics of Computing : Introduction to IT and Computer Ethics	Udemy	https://www.udemy.com/course/computer-ethics-introduction-to-computing-systems-ethics/
3.	Military Ethics: an Introduction	Future learn	https://www.futurelearn.com/courses/military-ethics
4.	Corporate Social Responsibility (CSR): A Strategic Approach	edX University of Pennsylvania	https://www.edx.org/course/corporate-social-responsibility-csr-a-strategic-aptm

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14. REAL TIME APPLICATIONS IN DAY TO DAY LIFE AND TO INDUSTRY
UNIT 5

Ethical Challenges Of Multinational Companies In Modern Business Management

❖The different ethical Issues faced by the Wal-Mart. MNC is clearly explained in the given link

https://www.ukessays.com/essays/management/ethical-challenges-of-multinational-companies-in-modern-business-management-essay.php

A Day in the Life of an Engineering Manager

- For most engineers, a typical work day might look like the following: coffee, code, coffee, code, lunch. After a few years, one might hope to move up the career ladder. And for some engineers, a position in management could be the next rung.
- But engineering managers aren't just senior-level coders or developers. There are a multitude of other skills necessary for being a successful people manager, including soft skills like clear communication and conflict resolution. So how do engineers learn to cultivate traits like empathy and mentorship in a tech-heavy job?

Few experiences of engineers as managers in their daily life is shared in the following links

https://www.karllhughes.com/posts/engineering-manager

https://www.karllhughes.com/posts/engineering-manager

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15. CONTENTS BEYOND SYLLABUS
UNIT 5

COMPUTER CRIMES

Most of the cyber criminals are anti-social element who gets motivated by the prior worst personal experience resulting in the crimes. Computer crimes can result in financial loss, information loss. Some major computer crimes in info space and as listed below which affects both work and personal life. (Herold, 2006)

- Hacking
- Virus
- Fraud
- Identity theft
- Denial Of Services – DoS
- Child pornography
- IRC crime
- Software piracy

Hacking

Computer hacking has become a status among many youngster.. Hackers are tend to have better understanding of coding and break into computers and are aware of bypassing the security or can remain undiscovered in the network for a certain time without leaving footprints for the forensics by wiping out the log files. When we talk about the evil we should see the other side of the coin the virtuous which is ethical hacking .Ethical hacking is meant for attacking the computer system or network with the full permission of the sole. The most common ethical hacking technique is penetration test by inducing the code to the system and to be aware of the vulnerabilities by negligence as the hackers look for loop holes and create back door. Many corporate create their own IT security policies to secure their network.

Virus

Computer viruses are in the top most list of threat faced by the Information world. Computer program created to do nasty works inside which enters into a computer by mean of downloading softwares or mp3 files which calls them as "Free softwares". Worm is much similar to the virus which gets replicated by human intervention; until and unless provoked remain quiet in the corner of the system. Trojan horses are interested in creating a back-door for the intruders in a compromised machine. Deploying a powerful anti-virus with the latest released virus definitions can help us in leading a peaceful life (Williams, 2003)

Identity theft

Financial identity theft is the widely asserted crime happening in the Internet. These intruders get into the system pretending the account holder of the bank by inputting false identity. A criminal identity thief uses the personal information including the national insurance number and social security number to do crime and leave it as the proof of identity. Medical identity theft is done by using another person's identity to obtain medical care or drugs where law restricts to purchase drugs. Computer security is the ultimate solution for identity theft by sanitizing the obsolete data in system and spam mails by installing anti-virus software which has powerful tools to protect online transactions. Most of the nations have law in place against these kind of identity thefts which includes heavy financial fine and imprisonment.

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Denial of Services-DoS

Denial of service won't affect neither the system security or destruct information loss any files make financial theft but can affect the targeted business by putting on fire ranging from second to days by making unavailability of a critical server, email or website which provides service as business . A buffer overflow with a prior assumption that of the weakness of programming or architecture of the applications used are the famous denial of service attacks that bench marked the history so far. Disabling the ICMP traffic in the firewall can prevent the ping of death. Implementing strong encryption techniques and deploying domain naming service. To overcome DoS attacks are by knowing the infrastructure and regular testing for complexity, keeping spam filters will limit mailbox getting flooded, if the business demands buying a space in the cloud or by keeping dramatically huge processor power and server volume will help to prevent such kind of issues.

Pornography

Pornography refers to any kind of image displaying a person who is being engaged in explicit sexual activity. In countries like Spain child pornography is a criminal offense if conventional pornography was in the paper based hard copy or video tape as technology advanced the Internet based pornography is image files or video files and on-line buffering. Severely punishing these criminals and these punishments to be brought to the notice of general public for their awareness that it will their experience if they do so. Association of sites advocating child protection ASACP was found in 1996 is meant for reporting child abuses in the society anonymously aiming to catch hold of the pornographers, they have a fully functional hot line and website asacp.org where people are report the crime . The RTA certificate helps children from accessing pornographic websites which is available Internet wide free of cost. (Barron's Educational Series, Inc, 1999)

IRC Crimes

Many people lose their privacy when then enter the IRC (Internet Relay Chat) rooms. The chat rooms become the playground of cyber criminals, hackers and terrorists. Verbal IRC crimes are where people get abused verbally in the chat rooms. The second category is clone attack where randomly synchronized codes are generated on behalf of usernames by robots to generate a random character which gets multiplied by the per-written code result in flooding and the chat server gets freezes due to Denial of service. Criminals use Cyber stalking to harass women and celebrities. IRC crimes can be stopped by simply ignoring or banning the user id which almost all chat sites allow.

Salami Attack

All these times we were talking about a crime which is relevant only to single attack , now lets talk about a series of tiny attacks which is categorized as salami slicing in computer cyber terms. The attacker steals a comparatively small amount of money from the account and the actual account holder will assume that it might be a service charge of any bank transactions or transaction which they might have done during shopping and won't be bothered to scrutinise the actual fact. such attacks will be done will multiple users which is like an atomic reaction.

Botnet

A bot is defined as zombie army which is an interconnected network of computers on the Internet with malicious infected software robots. The zombies army can be controlled only by the owner of the respective program. Botnet which is a malware can result in Denial of service attack. Criminals use this technique to utilize the shared processing power and memory Gradually the users system will face slow down. The solution to overcome the botnet is to install antivirus which can detect the malwares .(Parker, 1999)

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16. Assessment Schedule

Assessment	Proposed Date	Actual Date
Unit 1 Assignment Assessment		
Unit Test 1		
Unit 2 Assignment Assessment		
Internal Assessment 1		
Retest for IA 1		
Unit 3 Assignment Assessment		
Unit Test 2		
Unit 4 Assignment Assessment		
Internal Assessment 2		
Retest for IA 2		
Unit 5 Assignment Assessment		
Revision Test 1		
Revision Test 2		
Model Exam		
Remodel Exam		
University Exam		

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17.PRESCRIBED TEXT BOOKS & REFERENCE BOOKS

TEXT BOOKS:

1.Mike W. Martin and Roland Schinzinger, —Ethics in Engineering||, Tata McGraw Hill, New Delhi, 2003.

2.Govindarajan M, Natarajan S, Senthil Kumar V. S, —Engineering Ethics||, Prentice Hall of India, New Delhi, 2004.

REFERENCES:

1.Charles B. Fleddermann, —Engineering Ethics||, Pearson Prentice Hall, New Jersey, 2004.

2. Charles E. Harris, Michael S. Pritchard and Michael J. Rabins, —Engineering Ethics – Concepts and Cases||, Cengage Learning, 2009.

3.John R Boatright, —Ethics and the Conduct of Business||, Pearson Education, New Delhi, 2003

4.Edmund G Seebauer and Robert L Barry, —Fundamentals of Ethics for Scientists and Engineers||, Oxford University Press, Oxford, 2001.

5.Laura P. Hartman and Joe Desjardins, —Business Ethics: Decision Making for Personal Integrity and Social Responsibility|| Mc Graw Hill education, India Pvt.Ltd.,New Delhi, 2013.

6.World Community Service Centre, _ Value Education', Vethathiri publications, Erode, 2011.

Web sources:

1. www.onlineethics.org
2. www.nspe.org
3. www.globalethics.org
4. www.ethics.org

18. MINI PROJECT SUGGESTIONS

1.Case Study on Global Business Ethics

Issue of Price Cutting

- The increased competition among the Cement dealers in Kerala (India) has resulted in many of them selling cement at a lesser price than the stipulated market price. The intense price cutting has been worrying the cement companies and the KCDA (Kerala Cement Dealers Association) as it negatively affects the market price of cement and also the profit margin of the dealers. This has also resulted in unhealthy competition among the dealers. After many discussions and trials, the KCDA and the cement companies have arrived at a conclusion that they'll suspend the dealers, for a stipulated period, if they are caught in price cutting. This decision has been able to bring things under control to a certain extent.
- However, the rule breakers have started finding their way out to beat the system. They do not mind running the risk of a suspension as the punishments currently in place are not stringent enough. Some of the dealers use false proofs with which, the chances of getting caught are bleak. Some of them negotiate with the customers and promise them to give them the cement for a lesser price, provided they keep it a secret. Some of them promise the customers to give huge discounts on other items they buy. Due to many such issues, which are beyond control, the company and the association have not been able to bring in a fool-proof system. This is making the life very difficult for dealers who are ethical. Furthermore, the rule breakers end up making undeserved profits. Hopelessness in finding a fool-proof solution has forced some dealers who are ethical to take up unethical ways.

Share your views to solve this issue

2. Case study on ethics for computer and software engineers

A Breach in Security

- Data-Time Inc. is a company which manages databases for a large city in Colorado. Included in these databases is information collected from the city's homeless shelters and free clinics. Specifically, the databases contain personal information of the users of these services over the past 10 years; this includes people's Social Security numbers and health records.
- This data is highly secure and only accessible to the employees of Data-Time Inc. Employees are given a laptop when they are hired which allows them to access the database remotely. Unfortunately, one of these laptops is stolen and the security of the database is compromised.
- A majority of the people whose information was compromised are homeless; therefore there is no easy way to contact them in order to alert them of the security breach. How should Data-Time Inc. manage this breach in security?

Share your views to solve this issue

Thank you

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